SCC Branch Book

Structural Counterfactual Capacity, Minimal Carrier, and Endpoint Admissibility

Elements of Nested Fibrational Cosmology

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Front Block

Imported spine results.

(1) Book I: Observational Quotient Theorem (Thm. I.6.1); Collapse Gates CG-1–CG-6 (SRs I.1.4–I.1.9); Admissible-Class Canonicity (Thm. I.3.6); Behavioral Congruence Canonicity (Lem. I.3.5).
(2) Book II: Boundary Stabilization Theorem (Thm. II.8.1); Phase-A/B/C classification (Def. II.5.1); PASS as Derived Necessity (Thm. II.5.14); Binary Rigidity (Thm. II.5.8).
(3) Book III: Unified Coercive-Transport Inequality (Thm. III.5.1); Depth-Sum Convergence (Thm. III.5.4).
(4) Book IV: Universal Source Theorem (Thm. IV.7.1); five source-side rungs ORA–UC (Thms. IV.2.1–IV.3.1).
(5) Book V: Branch Projection Theorem (Prop. V.3.1); UBLT (Thm. V.7.4); Certified Branch Legitimacy Theorem (Thm. V.8.1).
(6) Book VII: Canonical Closure Schema Theorem (Thm. VII.6.4); No-Hidden-Upgrade Rule (Prop. VII.6.x); Non-Retroactivity (Thm. VII.6.y).

Branch-specific data.

(1) Declared SCC target category: the weak structural witness category $\mathcal{C}_{\mathrm{SCC}}$, whose objects are SCC quadruples (A_X, W_X, T_X, D_X) (support skeleton, witness family, transported invariant, defect ledger).
(2) The SCC terminal predicate τ_{SCC}: the structural counterfactual capacity condition on an SCC object.
(3) The SCC dependency spine: ORA $\Rightarrow$ CTM $\Rightarrow$ TIN $\Rightarrow$ MCS $\Rightarrow$ WeakGlue $\Rightarrow$ UC $\Rightarrow$ TSI.
(4) Four open obligations: O-SCC.MCS (main frontier), O-SCC.WeakGlue (prerequisite to MCS), O-SCC.UC, O-SCC.TSI.

Endpoint target. Persistence-selected structural counterfactual capacity: the certified structural property of an NFC-source-descended object that supports lawful counterfactual reasoning about its own observer-relative distinguished states, under the constraints of the canonical source image. No phenomenological, conscious, or experiential content is asserted.

Bridge stack.

(1) Licensed source-descended realization via F_{SCC} (ORA–UC rungs from Book IV).
(2) SCC Minimal Structural Carrier Selection (O-SCC.MCS): the main open burden.
(3) SCC Branch-Local Pairwise Amalgamation (O-SCC.WeakGlue): prerequisite to MCS discharge.
(4) Branchwise SCC completion $\mathcal{U}_{\mathrm{SCC}}$ (O-SCC.UC).
(5) SCC Target/Structural Identification (O-SCC.TSI): full endpoint admissibility beyond structural-label form.

Current status. **CERT-PROJ at most. Main frontier: O-SCC.MCS.**

The SCC branch is a lawful NFC descendant. All four core obligations (MCS, WeakGlue, UC, TSI) are open in canonical papers. The branch is at CERT-PROJ —

licensed projection at stated structural scope — and cannot advance to CERT-CLOSE until the MCS theorem is canonically proved.

SCC.0 Purpose

This branch studies the structural counterfactual capacity of NFC-source-descended objects. Its task: determine whether the canonical spine produces a certified object that supports lawful counterfactual reasoning about its own observer-relative distinguished states, under the no-smuggling discipline.

The SCC branch occupies a unique position in the NFC program. Its endpoint is the furthest from direct physical measurement and the most philosophically sensitive. The program's response to this is not to retreat from the endpoint but to impose the strictest possible labeling discipline: the endpoint label "structural counterfactual capacity" is the strongest admissible label consistent with the spine's theorems, and no phenomenological or ontological upgrade is licensed.

The canonical discipline has one specific consequence: the word "consciousness" and its cognates are not licensed at any stage in the canonical theorem flow. This is not a philosophical stance on whether consciousness is real; it is a status-honesty claim that the current canonical framework has not proved a consciousness theorem and does not use consciousness as a premise.

1. Branch Governance Preamble

Standing Rule 1.1 ([D]). (SCC Anti-Smuggling.) No target-side label for the SCC branch may introduce content beyond the certified source image. In particular: no phenomenology, no independent ontology, no private experiential primitive, no consciousness claim, and no import from empirical cognitive science may enter the canonical theorem flow without explicit declaration and separate canonical proof.

Standing Rule 1.2 ([D]). (Consciousness Non-Claim.) No theorem in this book licenses the label "consciousness," "subjective experience," "qualia," "phenomenal states," or any equivalent label. The strongest admissible endpoint label is *persistence-selected structural counterfactual capacity.* Any stronger label requires a separate canonical proof not present in this book.

Proposition 1.3 ([U]). (Formation.) *The SCC branch candidate* $\mathfrak{B}_{\mathrm{SCC}}$ *is constitutionally well-formed: all five components* ($\mathbf{U}$, $\mathcal{C}_{\mathrm{SCC}}$, $F_{\mathrm{SCC}}, \mathrm{Cert}_{\mathrm{SCC}}$, $\tau_{\mathrm{SCC}}, \mathrm{FF}_{\mathrm{SCC}}$) *are declared.*

Proof. All five components are declared in the front block and defined in Sections 3–5. By the Certified Branch Legitimacy Theorem (Book V, Thm. V.8.1), the branch is constitutionally formed. □

Proposition 1.4 ([U]). (Interpretive Label Admissibility.) *The label "persistence-selected structural counterfactual capacity" passes all four tests of the No-Hidden-Upgrade Rule (Book VII, Prop. VII.6.x): (i) it is eliminable (reduces to the structural predicate* τ_{SCC}*); (ii) it asserts no independent ontology; (iii) it imports no phenomenology; (iv) it does not depend on data outside the certified source image.*

Proof. Test (i): the label is eliminable in favor of the structural predicate on the SCC quadruple (A_X, W_X, T_X, D_X). Test (ii): SCC is a structural property of NFC configurations; no independent mental ontology is asserted. Test (iii): no experiential or phenomenal content enters the definition of τ_{SCC}. Test (iv): the predicate is computed entirely from the certified source image $F_{\mathrm{SCC}}(\mathbf{U})$. □

Theorem 1.5 ([U]). (SCC TSI Firewall.) *A proposed endpoint label L is admissible for the SCC branch if and only if none of the four interpretive failures occur:*

(i) *FI-IADD: L adds theorem content not in the certified spine image.*
(ii) *FI-IONT: L asserts an independent ontology.*
(iii) *FI-IPHEN: L imports phenomenology absent from the theorem content.*
(iv) *FI-ISRC: L depends on data outside the certified source image.*

The label "persistence-selected structural counterfactual capacity" passes all four tests. "Consciousness" and its cognates fail FI-IONT and FI-IPHEN at the current canonical stage.

Proof. The SCC TSI firewall is the No-Hidden-Upgrade Rule (Book VII, Prop. VII.6.x) applied to the SCC endpoint. Tests (i)–(iv) are the exact four tests of that rule. For the admissible label: Proposition 1.4. For consciousness: at the current stage, no theorem in the canonical spine or this branch book identifies the SCC structural predicate with phenomenal consciousness; importing that identification would constitute FI-IONT (independent mental ontology) and FI-IPHEN (phenomenology not in theorem content). □

Remark 1.6 ([D]). (Status vocabulary.) Standard FI-code and positive status alphabet applies. Current status: CERT-PROJ at most. No claim in this book exceeds CERT-PROJ.

2. Imported Spine Structure

Imports from Book I. The observational quotient Σ_∞ and defect ledger Δ are the primitive inputs to the SCC carrier reduction program. Admissible-Class Canonicity (Thm. I.3.6) ensures $E_{\max}$ is uniquely defined: the SCC minimal carrier is a carrier within this maximal class.

Imports from Book II. PASS (Thm. II.5.14) governs the SCC branch: a persistence-selected object satisfying PASS is the precise domain of the SCC terminal predicate. Binary Rigidity (Thm. II.5.8) provides the fiber structure that the SCC carrier reduction operates on: each non-trivial stable fiber has cardinality exactly 2, so the SCC carrier space is stratified into binary layers.

Imports from Book III. The Unified Coercive-Transport Inequality (Thm. III.5.1) and Depth-Sum Convergence (Thm. III.5.4) provide the transport control for SCC data across admissible enlargements. The SCC branch uses Θ_n in the full range $[0, 1]$, depending on the coercivity regime of the specific SCC realization.

Imports from Books IV–VII. Universal source descent (Book IV), branch legitimacy (Book V), force-governance and No-Hidden-Upgrade Rule (Book VII).

3. Branch-Specific Arena

DEFINITION 3.1 ([D]). An *SCC object* is a quadruple $X = (\mathcal{A}_X, \mathcal{W}_X, \mathcal{T}_X, \mathcal{D}_X)$ where:

(1) $\mathcal{A}_X$ is the *support skeleton*: the persistence-selected admissible sub-configuration descended from the NFC source.
(2) $\mathcal{W}_X$ is the *witness family*: the finite collection of admissible local tests that certify the SCC structural property.
(3) $\mathcal{T}_X$ is the *transported invariant*: the structural quantity preserved under admissible enlargement.
(4) $\mathcal{D}_X$ is the *defect/interface ledger*: the cumulative defect record of the SCC object's continuation history.

DEFINITION 3.2 ([D]). The *SCC target category* $\mathcal{C}_{\mathrm{SCC}}$ is the weak structural witness category whose objects are SCC objects (Definition 3.1) and whose morphisms are admissible SCC enlargements (transport-compatible admissible extensions preserving $\mathcal{T}_X$).

REMARK 3.3 ([R]). $\mathcal{C}_{\mathrm{SCC}}$ is a *structural* category. Its objects are not minds, agents, or experiential subjects. They are NFC-source-descended configurations equipped with a certified structural predicate. The "counterfactual capacity" in the endpoint label refers to the configuration's structural ability to support counterfactual relational distinctions under the admissible test family — not to any capacity for reasoning or experience.

DEFINITION 3.4 ([D]). The *SCC terminal predicate* $\tau_{\mathrm{SCC}}(X)$ holds of an SCC object X if:

(1) X is persistence-selected: it has survived all six collapse gates (Book I, SRs I.1.4–I.1.9) and satisfies PASS (Book II, Thm. II.5.14).
(2) X admits a minimal structural carrier (MCS): a minimal faithful support family for the transported invariant $\mathcal{T}_X$ (O-SCC.MCS, when discharged).
(3) $\mathcal{T}_X$ is counterfactually distinguishing: the transported invariant separates distinct admissible histories at the structural level.

DEFINITION 3.5 ([D]). The *SCC realization functor* $F_{\mathrm{SCC}} : \mathrm{POT} \to \mathcal{C}_{\mathrm{SCC}}$ maps each POT object P to the SCC object $F_{\mathrm{SCC}}(P) = (\mathcal{A}_P, \mathcal{W}_P, \mathcal{T}_P, \mathcal{D}_P)$ carrying the collar-inherited SCC structural data.

DEFINITION 3.6 ([D]). The *recursively stable maximal carrier* $\mathrm{CapRec}(X) = \mathfrak{C}^{\mathrm{rec}}_{\max}(X)$ is the carrier-equivalence class of X stable under all admissible continuation that do not introduce new defect events. It defines the branch-identity class of the SCC object under the equivalence relation $\sim_{\mathrm{cap}}$ (cap-equivalence).

4. Lawful Descent and Branch Identity

PROPOSITION 4.1 ([U]). (SCC Branch is Lawful.) *The branch candidate* $\mathfrak{B}_{\mathrm{SCC}}$ *is a lawful NFC branch satisfying all five conditions of the Certified Branch Legitimacy Theorem (Book V, Thm. V.8.1).*

Proof. (1) Source descent: F_{SCC} descends from $\mathbf{U}$ via Book IV, with ORA–UC rungs imported as discharged [U] results. (2) Intake survival: the ORA–CTM–TIN rungs are proposed/partial; MCS is the main open burden. The branch survives through TIN at its current scope. (3) Ledger completeness: O-SCC.MCS through O-SCC.TSI are explicitly ledgered in Section 5. (4) Status honesty: no claim exceeds CERT-PROJ. (5) Non-retroactivity: the SCC branch does not affect YM, NS, or GR (Book VII, Thm. VII.6.y). □

Proposition 4.2 ([U]). (Full Distinctness.) *The SCC branch is genuinely distinct from the YM, NS, and GR branches. Its terminal predicate (persistence-selected structural counterfactual capacity of the transported invariant) is not the terminal predicate of any other named branch.*

Proof. YM targets a spectral mass gap; NS targets fluid regularity; GR targets Einstein-type structural compatibility of a metric field. None is the SCC terminal predicate. □

Proposition 4.3 ([U]). (Branch Trichotomy.) *Every recursively stabilized pair (X, Y) of SCC objects falls into exactly one of:*

(i) Collapse: $X \sim_{\mathrm{path}} Y$ *(path-equivalent under recursive stabilization; same branch identity class).*
(ii) Confluent distinctness: $X \not\sim_{\mathrm{path}} Y$ *but* $X \sim_{\mathrm{cap}} Y$ *(cap-confluent; distinct histories, same structural carrier class).*
(iii) Genuine branch distinctness: $X \not\sim_{\mathrm{path}} Y$ *and* $X \not\sim_{\mathrm{cap}} Y$ *(genuinely distinct SCC objects).*

Proof. The trichotomy follows from the definitions of $\sim_{\mathrm{path}}$ and $\sim_{\mathrm{cap}}$ as equivalence relations on the set of recursively stabilized SCC objects. Any two objects are either (i) path-equivalent, (ii) cap-confluent but not path-equivalent, or (iii) neither. These cases are mutually exclusive and exhaustive. □

Proposition 4.4 ([U]). (SCC Anti-Smuggling.) *No target-side label for the SCC branch may introduce content beyond the certified source image. Formally: for any proposed SCC datum d, if d requires data outside $F_{\mathrm{SCC}}(\mathbf{U})$, then d is prohibited as a carrier element.*

Proof. By Book I, Standing Rule I.1.3 (no-smuggling at the primitive level) and the branch-level extension of this rule (SR 1.1): any datum outside the certified source image is a silent import and violates the intake law (Book V, UBLT Thm. V.7.4, condition (4)). It is therefore excluded from the SCC carrier by the anti-smuggling gate. □

5. The Closure Ledger

ID	Name	St.	Blocks
O-SCC.WeakGlue	SCC Branch-Local Pairwise Amalgamation: pairwise amalgamation for SCC carrier data (branch-specific; distinct from source-side O-WeakGluing)	[C]	O-SCC.MCS (prerequisite)
O-SCC.MCS	SCC Minimal Structural Carrier Theorem: existence and uniqueness of the minimal faithful support family for $\mathcal{T}_X$	[O]	O-SCC.UC; CERT-PROJ upgrade
O-SCC.UC	Branchwise SCC Universal Completion: local SCC realizations factor through a single $\mathcal{U}_{\mathrm{SCC}}$	[O]	O-SCC.TSI
O-SCC.TSI	SCC Target/Structural Identification: full endpoint admissibility beyond structural-label form; counterfactual capacity certified at full TSI scope	[O]	CERT-CLOSE

Remark 5.1 ([R]). The dependency chain is: WeakGlue $\Rightarrow$ MCS $\Rightarrow$ UC $\Rightarrow$ TSI $\Rightarrow$ CERT-CLOSE. O-SCC.MCS is the primary failure frontier. O-SCC.WeakGlue has been conditionally discharged (see Theorems 5.2–5.6 below): its proof is conditional on the frozen Support Sufficiency Schema and the SCC object definitions, both of which are established in this book. O-SCC.MCS-Uniqueness follows (Theorem 5.8); O-SCC.MCS-Existence remains open and reduces to SCC-MinExist (5.11).

Theorem 5.2 ([C]). (SCC-W1: Witness Sufficiency Preserved Under Overlap.) *Let K and L be lawful SCC carrier families for the same SCC observable τ_{SCC}, with the same normalized witness-type support and the same transported persistence/defect-history support, and with realized-support data compatible with the same ORA/CTM/TIN/ledger structures. Let $K \wedge L$ denote their regimewise common realized-support subfamily. Then every witness/outcome support component genuinely required for τ_{SCC} is present in $K \wedge L$.*

Proof. Fix a regime R. Let w be a realized witness/outcome support component genuinely required for τ_{SCC} at R. Because K and L support the same SCC observable, the required witness/outcome support is the same in both after admissible TIN-compatible normalization. Suppose $w \notin K \wedge L$; then w fails to appear in at least

one of K_R, L_R as common realized support. Either that carrier does not support τ_{SCC} (contradicting same-observable), or w is replaced by unlicensed target-side enrichment (violating anti-smuggling, Book V). Both cases are excluded. Hence $w \in K \wedge L$. □

THEOREM 5.3 ([C]). (SCC-W2: Normalized Witness-Type Support Preserved.) *Under the hypotheses of Theorem 5.2, every normalized witness-type class genuinely required to interpret τ_{SCC} survives into $K \wedge L$.*

PROOF. Normalized witness-type classes are realized after admissible SCC normalization. Sameness of normalized witness-type support is assumed. If a required class c were absent from $K \wedge L$, then after admissible normalization one carrier would fail to share a required normalized witness-type class with the other, contradicting same-normalized-type-support. Alternatively, c would be redundant presentation-level surplus, which is excluded by the realized-support order. Hence $c \in K \wedge L$. □

THEOREM 5.4 ([C]). (SCC-W3: Transported Persistence/Defect-History Support Preserved.) *Under the hypotheses of Theorem 5.2, every transported persistence/defect-history support component genuinely required for τ_{SCC} survives into $K \wedge L$.*

PROOF. Transported defect/history support is defined along a shared ORA/CTM transport spine up to target equivalence and ledger visibility. If a required component d were absent from $K \wedge L$, one carrier would fail to share it (contradicting same-transported-support), or it would be replaced by unlicensed enrichment (violating anti-smuggling). Both cases are excluded. Hence $d \in K \wedge L$. □

THEOREM 5.5 ([C]). (SCC-W4: ORA/CTM/TIN/Ledger Compatibility Preserved.) *Under the hypotheses of Theorem 5.2, the common realized-support carrier $K \wedge L$ inherits ORA/CTM/TIN/ledger compatibility.*

PROOF. $K \wedge L$ is formed by intersecting realized support, not raw presentation data. Each retained support component was already ORA-admissible, CTM-transport-compatible, TIN-normalized, and ledger-consistent in both K and L. (i) *ORA*: restriction to common realized support introduces no new restriction datum, so ORA status is unchanged. (ii) *CTM*: the shared transport spine already satisfies commutativity in both carriers; restriction removes branches without breaking commutativity. (iii) *TIN*: normalization was already agreed after TIN-compatible normalization, so no fresh ambiguity arises. (iv) *Ledger*: retained defects were ledger-visible in both; anti-smuggling blocks hidden replacement. Hence all four compatibility structures are inherited. □

THEOREM 5.6 ([C]). (O-SCC.WeakGlue: SCC Pairwise Amalgamation.) *If two lawful SCC carrier families K, L support the same SCC observable τ_{SCC}, have the same normalized witness-type support, the same transported persistence/defect-history support, and are ORA/CTM/TIN/ledger compatible, then their common realized-support carrier $K \wedge L$ is lawful and faithful.*

PROOF. By Theorems 5.2–5.5, all four support layers and all four compatibility structures survive passage to $K \wedge L$. Therefore $K \wedge L$ satisfies all conditions for a lawful SCC carrier. It is faithful because all support components required for τ_{SCC} are present (Theorem 5.2). □

THEOREM 5.7 ([C]). (SCC Shared-Support Faithfulness.) *If K and L are lawful faithful SCC carrier families for the same τ_{SCC}, then their common realized-support carrier $K \wedge L$ is also lawful and faithful.*

PROOF. Immediate from Theorem 5.6. □

THEOREM 5.8 ([C]). (SCC-MCS Uniqueness.) *If a minimal lawful faithful SCC carrier exists, it is unique up to carrier equivalence.*

PROOF. Let K and L be two minimal lawful faithful SCC carrier families for the same τ_{SCC}. By Theorem 5.6, $M := K \wedge L$ is lawful and faithful. By construction $M \preceq K$ and $M \preceq L$ in the realized-support order. Minimality of K and L forces $K = M$ and $L = M$, hence $K \sim L$. □

REMARK 5.9 ([R]). Theorem 5.8 discharges the *uniqueness* half of O-SCC.MCS. The *existence* half — whether a minimal element exists in the carrier order — remains open as O-SCC.MCS (5.11 below). Existence reduces to the finite-localizability of the transported invariant $\mathcal{T}_X$; see the SCC-FL program in the Speculative Holding. This remark is governed by Standing Rule **??**.

OPEN OBLIGATION 5.10 ([O]). (O-SCC.WeakGlue: DISCHARGED.) See Theorem 5.6. This obligation is now [C].

OPEN OBLIGATION 5.11 ([O]). (O-SCC.MCS: Minimal Structural Carrier Theorem.) For every SCC object X with transported invariant $\mathcal{T}_X$, there exists a minimal faithful support family $\mathcal{M}_X \subseteq \mathcal{W}_X$ such that:

(1) $\mathcal{M}_X$ is a support family for $\mathcal{T}_X$: $\mathcal{T}_X$ is recoverable from $\mathcal{M}_X$-level data alone.
(2) $\mathcal{M}_X$ is minimal: no proper sub-family of $\mathcal{M}_X$ supports $\mathcal{T}_X$.
(3) $\mathcal{M}_X$ is unique up to canonical carrier equivalence.

Status: Uniqueness is now proved (Theorem 5.8). Existence is the remaining open question, and reduces to proving that every descending chain of lawful faithful SCC carriers has a lawful faithful lower bound (SCC-MinExist). Main open burden of the SCC branch.

OPEN OBLIGATION 5.12 ([O]). (O-SCC.UC: Branchwise SCC Completion.) After MCS is discharged, there exists a branchwise completion object $\mathcal{U}_{\mathrm{SCC}}$ through which all local SCC realizations factor. This is the SCC analogue of the Universal Completion Theorem (Book IV) at the branch level. $\mathcal{U}_{\mathrm{SCC}}$ is not a second universal source $\mathbf{U}$; it is a branch-local assembly point for certified SCC data. Its existence depends on O-SCC.MCS.

REMARK 5.13 ([R]). The dependency chain is: WeakGlue $\Rightarrow$ MCS $\Rightarrow$ UC $\Rightarrow$ TSI $\Rightarrow$ CERT-CLOSE. O-SCC.MCS is the primary failure frontier: it is the smallest theorem that would unlock the rest of the chain. The WeakGlue obligation is a prerequisite that must be discharged before MCS can be addressed.

OPEN OBLIGATION 5.14 ([O]). (O-SCC.UC: Branchwise SCC Completion.) After MCS is discharged, there exists a branchwise completion object $\mathcal{U}_{\mathrm{SCC}}$ through which all

local SCC realizations factor. This is the SCC analogue of the Universal Completion Theorem (Book IV, Thm. IV.3.1) at the branch level.

$\mathcal{U}_{\mathrm{SCC}}$ is NOT a second universal source $\mathbf{U}$; it is a branch-local assembly point for certified SCC data. Its existence depends on O-SCC.MCS.

Open Obligation 5.15 ([O]). (O-SCC.TSI: Target/Structural Identification.) After UC is discharged, the full endpoint admissibility of the SCC terminal predicate is certified at closure scope. The structural label "persistence-selected structural counterfactual capacity" is promoted to full TSI force.

Status: Structural label form only (weak form) at current stage. Full TSI requires MCS and UC discharge.

6. Named Failure Frontier

Theorem 6.1 ([U]). (SCC Failure Frontier.) *The SCC branch has a precisely named failure frontier:*

(i) ***Primary frontier (O-SCC.MCS):*** *the Minimal Structural Carrier Theorem. Until this is proved canonically, the SCC branch cannot advance beyond CERT-PROJ. Silent carrier inflation remains a structural risk.*
(ii) ***Prerequisite frontier (O-SCC.WeakGlue):*** *the SCC branch-local pairwise amalgamation lemma. Must be discharged before MCS can be addressed.*
(iii) ***Completion frontier (O-SCC.UC):*** *branchwise SCC universal completion. Blocked by MCS.*
(iv) ***Identification frontier (O-SCC.TSI):*** *full endpoint TSI certification. Blocked by UC.*

Proof. The Frontier Stability Theorem (Book VII, Thm. VII.4.3) requires a positive named failure frontier. The four items above are the SCC branch's frontier: each is a precise mathematical statement, currently unresolved in canonical papers, blocking a specific step in the closure chain. □

7. SCC ICDC: Scale-Limit Direct System

Theorem 7.1 ([C]). (SCC-ICDC: Scale-Limit BP-SCC Direct System.) [Conditional on O-SCC.MCS and O-SCC.WeakGlue discharged, PASS, Phase-A.] *Let* $(X_n)_{n\geq n_0}$ *be a Van Hove exhaustion of SCC objects with transported invariants* $(\mathcal{T}_{X_n})$ *satisfying the five ICDC conditions of Book III, Scholium III.8.1. Then the scale-limit SCC object* $X_\infty := \varinjlim_{\partial} X_n$ *exists and carries a coherent transported invariant* $\mathcal{T}_{X_\infty}$*, with the SCC structural property inherited from the finite stages.*

Proof sketch. Apply the ICDC Schema (Book III, Sch. III.8.1) with the SCC specialization $\Theta_n \in [0,1]$ (regime-dependent). The five ICDC conditions (one-step inheritance, interface localization, two-step transitivity, source-image discipline, summable residues) are verified for the SCC transported invariant using O-SCC.MCS (minimal carrier ensures faithful support survives the limit) and O-SCC.WeakGlue (pairwise amalgamation ensures the direct limit is well-defined). □

8. R1: SCC Branch-Projection

THEOREM 8.1 ([O]). (R1: SCC Branch-Projection, Weak Form.) [Reserved — no present theorem force.]

The SCC Branch-Projection Theorem in its weak form: there exists a canonical projection from the persistence-selected SCC sector to the space of structural counterfactual capacity labels, analogous to the Branch Projection Theorem (Book V, Prop. V.3.1) at the SCC endpoint level.

Status: *Reserved. This is the R1 obligation from the retired Branch A* governance layer. It cannot be cited as a discharged theorem. It is recorded here to make the SCC program's full obligation structure visible. Discharge requires MCS, UC, and TSI to be first established.*

9. Final Status Proposition

PROPOSITION 9.1 ([U]). (Canonical Status.) *The SCC branch currently satisfies:*

(1) ***Lawful branch:*** *confirmed (Prop. 4.1).*
(2) ***Anti-smuggling:*** *enforced at the primitive level (SR 1.1) and at the endpoint level (Thm. 1.5).*
(3) ***TSI:*** *structural-label form only (weak form). Full TSI blocked by O-SCC.MCS. Failure code FI-TSI.*
(4) ***Strongest admissible endpoint label:*** persistence-selected structural counterfactual capacity.
(5) ***What is NOT licensed:*** *consciousness, phenomenal experience, subjective states, qualia, or any stronger label. These fail the SCC TSI firewall (FI-IONT, FI-IPHEN).*
(6) ***Status:*** *CERT-PROJ at most. NOT CERT-CLOSE.*

10. Boundary of the SCC Branch Book

What this branch book establishes.

(1) The SCC branch is a lawful NFC branch (CERT-PROJ).
(2) The SCC TSI firewall (Thm. 1.5): a formal theorem governing endpoint label admissibility.
(3) The SCC anti-smuggling proposition (Prop. 4.4).
(4) Branch trichotomy (Prop. 4.3): the three-way classification of SCC object pairs.
(5) The four precisely named open obligations (Thm. 6.1).
(6) The SCC-ICDC conditional theorem (Thm. 7.1).
(7) The R1 SCC Branch-Projection theorem recorded as a Reserved obligation (Thm. 8.1).

What this branch book does not establish.

(1) Discharge of any core obligation (WeakGlue, MCS, UC, TSI) as a canonical theorem.
(2) CERT-CLOSE status.

(3) Consciousness, phenomenal experience, or any stronger label than "persistence-selected structural counterfactual capacity."
(4) A quantum theory of consciousness or any AI-specific claim.
(5) Any empirical cognitive science import.

11. Dependency Ledger

Theorem	St.	Depends on	Exports to
1.3 Formation	[U]	Bk. V Thm. V.8.1	4.1
1.4 Label Admissibility	[U]	Bk. VII Prop. VII.6.x	1.5
1.5 TSI Firewall	[U]	1.4; FI codes	Endpoint label discipline
4.1 SCC Branch Lawful	[U]	1.3; Bk. IV–V	All subsequent SCC theorems
4.2 Full Distinctness	[U]	Terminal predicates	Branch catalog
4.3 Branch Trichotomy	[U]	$\sim_{\text{path}}$, $\sim_{\text{cap}}$ definitions	Branch identity logic
4.4 Anti-Smuggling	[U]	Bk. I SR I.1.3; 1.1	Carrier reduction; O-SCC.MCS
6.1 Failure Frontier	[U]	All obligations named	External scrutiny; CERT-PROJ posture
7.1 SCC-ICDC	[C]	O-SCC.MCS + WeakGlue; Bk. III Sch. III.8.1	O-SCC.UC; scale-limit SCC
8.1 R1 (Reserved)	[O]	O-SCC.MCS + UC + TSI	Future SCC closure program
9.1 Canonical Status	[U]	All of SCC Branch Book	Program ledger

Scholium 11.1 ([R]). The SCC Branch Book is frontier-honest. The canonical program makes no claim about consciousness and does not use consciousness as a premise. What it does claim is precisely delineated by the SCC TSI firewall: persistence-selected structural counterfactual capacity. This is a mathematically tractable structural property that emerges from the NFC source-descended carrier structure. Whether this structural property bears any relation to phenomenal consciousness is a question this book explicitly does not address, because addressing it would require theorem content not present in the canonical spine. The firewall is not a philosophical evasion; it is the canonical program's honest statement of where its theorem-bearing force currently ends.